

Introduction

The **Urban Heat Island (UHI)** effect occurs when urban areas are significantly warmer than nearby rural areas due to materials like **concrete**, **asphalt**, and **rooftops** that absorb and store heat (MetLink, n.d.). These surfaces heat up during the day and slowly release heat at night, keeping cities warmer for longer periods of time. This leads to increased **energy use**, higher levels of **air pollution**, and greater **health risks** during extreme heat.

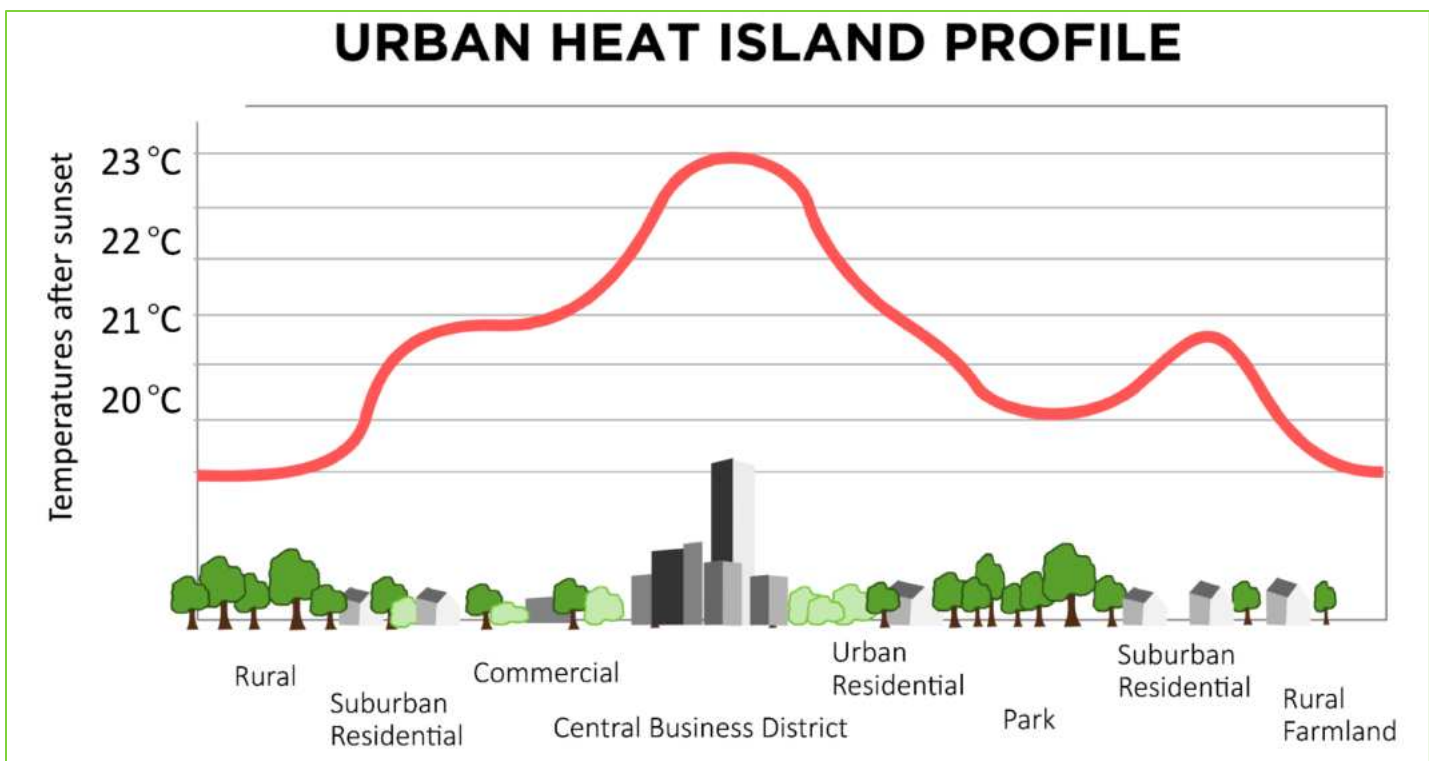


Figure 1. Urban Heat Island Profile

One possible solution to the Urban Heat Island effect is **radiative cooling**, which involves reflecting sunlight away from surfaces and releasing heat back into the atmosphere. **Reflective coatings**, often called **cool roof paints**, have been shown to reduce **surface temperatures** on a large scale. Less research has been done on small-scale applications using **low-cost** and accessible materials. The experiment tested whether an experimental paint mixture using reflective powders, including **barium sulfate** and **titanium dioxide**, could achieve similar cooling effects under controlled conditions.

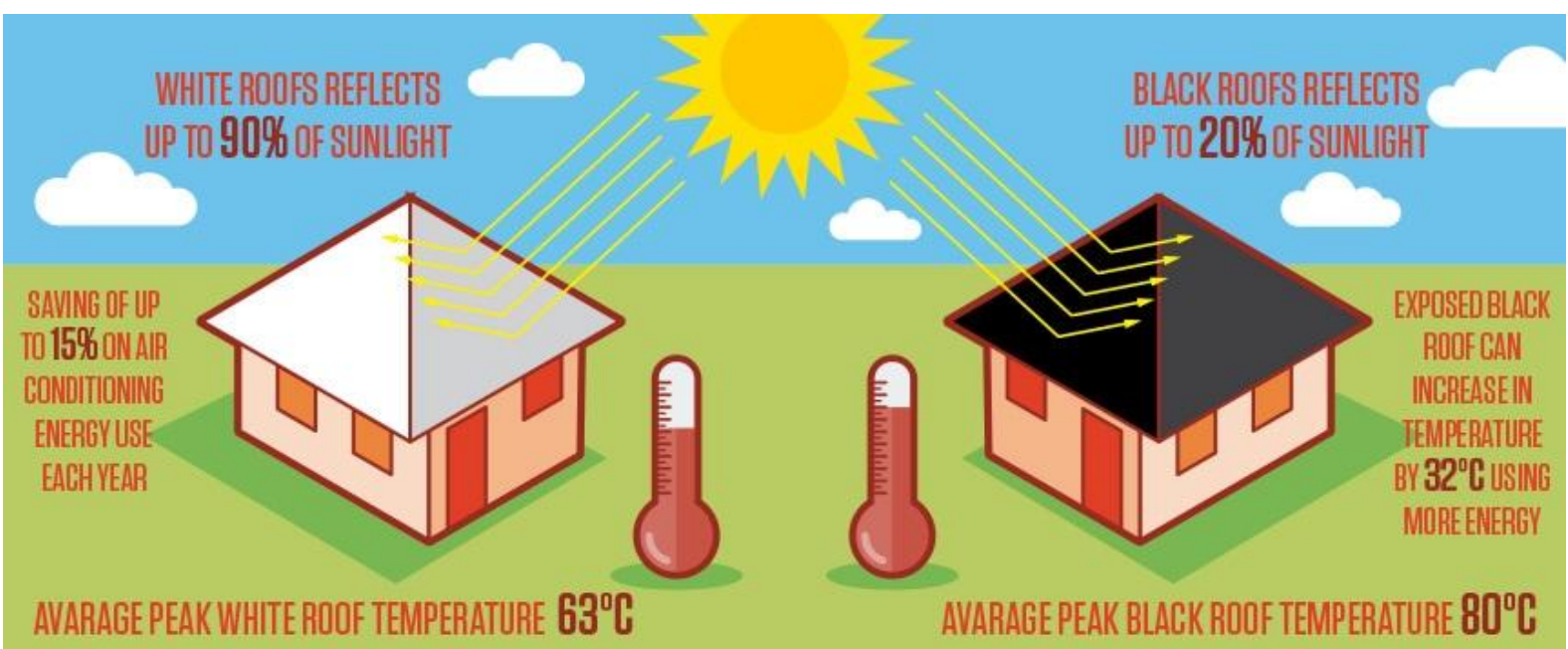


Figure 2. Comparison of Darker Cooled Roof vs. Lighter Cooled Roof

Purpose

Utilizing **reflective coatings** offers a potential strategy for reducing **surface heating** in urban environments. Urban areas experience elevated temperatures due to **heat-absorbing materials** such as concrete and asphalt, which contribute to the **Urban Heat Island (UHI)** effect. This effect increases energy demand, **worsens air quality**, and raises health risks during extreme heat events. Developing **low-cost reflective coatings** provides a practical approach to reducing **heat absorption** on common urban surfaces. This work focuses on evaluating whether **accessible reflective materials** can support temperature reduction strategies in built environments.

Developing a Coating Created With Barium Sulfate and Titanium Dioxide to Create a Cool Coat Solution For Concrete Buildings

Methods

Step 1

Barium Sulfate and Titanium Dioxide was mixed into white Low-VOC paint



Figure 3. Low VOC Paint Mixture

Step 2

Concrete brick samples were prepared in uniform sizes and labeled based on the paint type being tested.



Figure 4. Concrete brick sample

Step 3

Each brick was coated with either standard roof paint, reflective roof paint, or the experimental mixture, and all samples were allowed to fully dry.



Figure 5. Brick Coated with Roof Paint

Step 4

The bricks were placed under a halogen lamp at a consistent distance to simulate sunlight and ensure controlled conditions.



Figure 6. Halogen lamp + brick setup

Step 5

Surface temperatures were measured using an infrared thermometer before heating, immediately after heating, and after 20 minutes, and results were recorded for comparison.

Results

Paint Type	Brick #	Color?	Temp. Before (°F)	Temp. Immediately After (°F)	Temp. After 20 Mins (°F)
Reflective Roof Paint	#1	No	47.3	67.6	63.2
Reflective Roof Paint	#2	No	47.7	66.3	62.9
Regular Roof Paint	#1	No	47.6	71.8	67.5
Regular Roof Paint	#2	No	47.5	71.8	67.5
Low VOC Paint + Mixture	#1	No	47.4	65.8	61.9
Low VOC Paint + Mixture	#2	No	47.6	66.1	62.2

Figure 7. Data table with temp. results

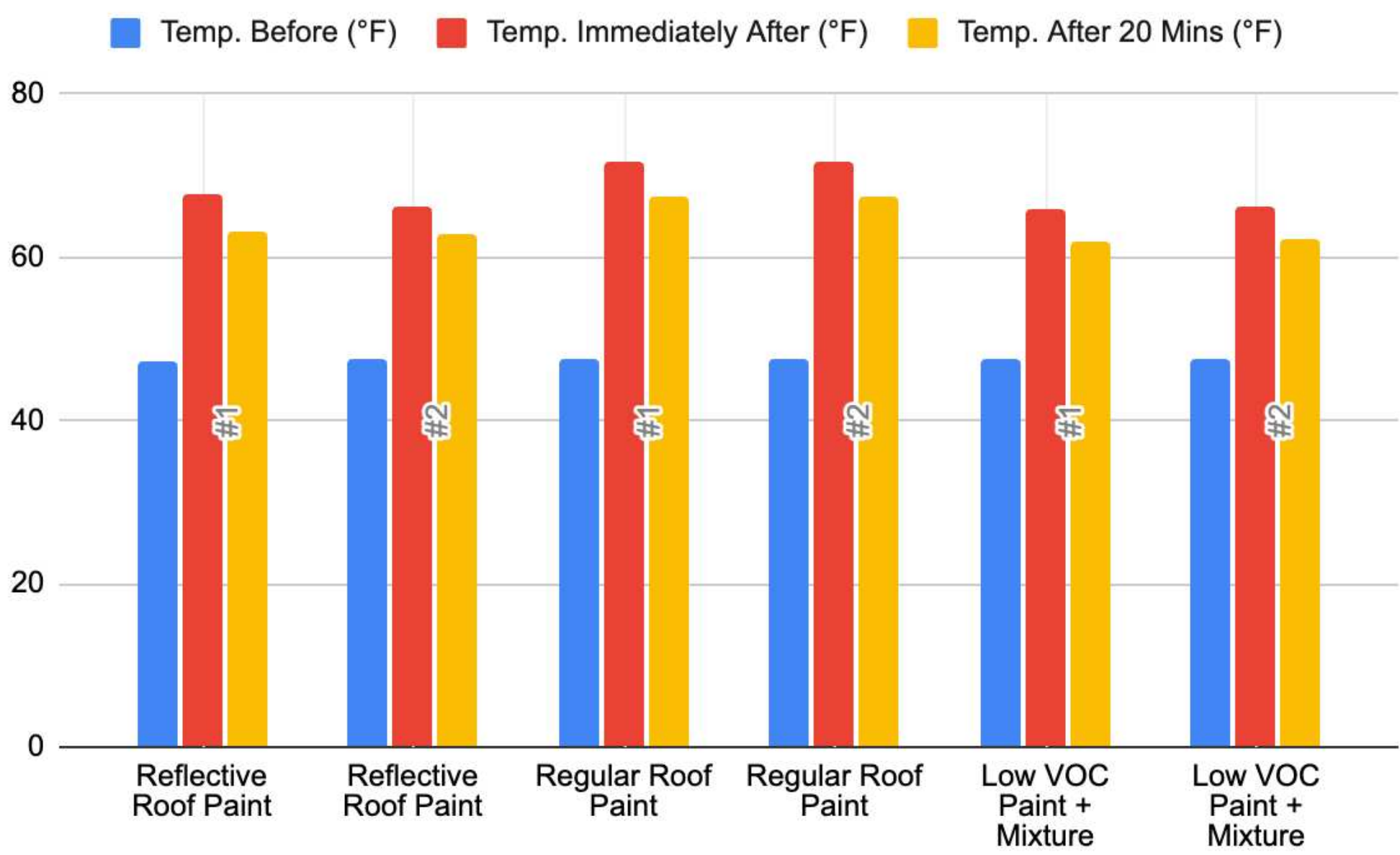


Figure 8. Data Graph with Temp. Results

Conclusion

The results show that **reflective coatings** are more effective at reducing surface temperature than standard roof paint. The reflective paint consistently reached **lower peak temperatures** and cooled more efficiently over time compared to the control. The experimental mixture showed **lower peak temperatures** than the control, but slightly higher temperatures and slower cooling compared to the commercial reflective coating, indicating **lower overall reflectivity** than the commercial product but improved performance over standard paint.

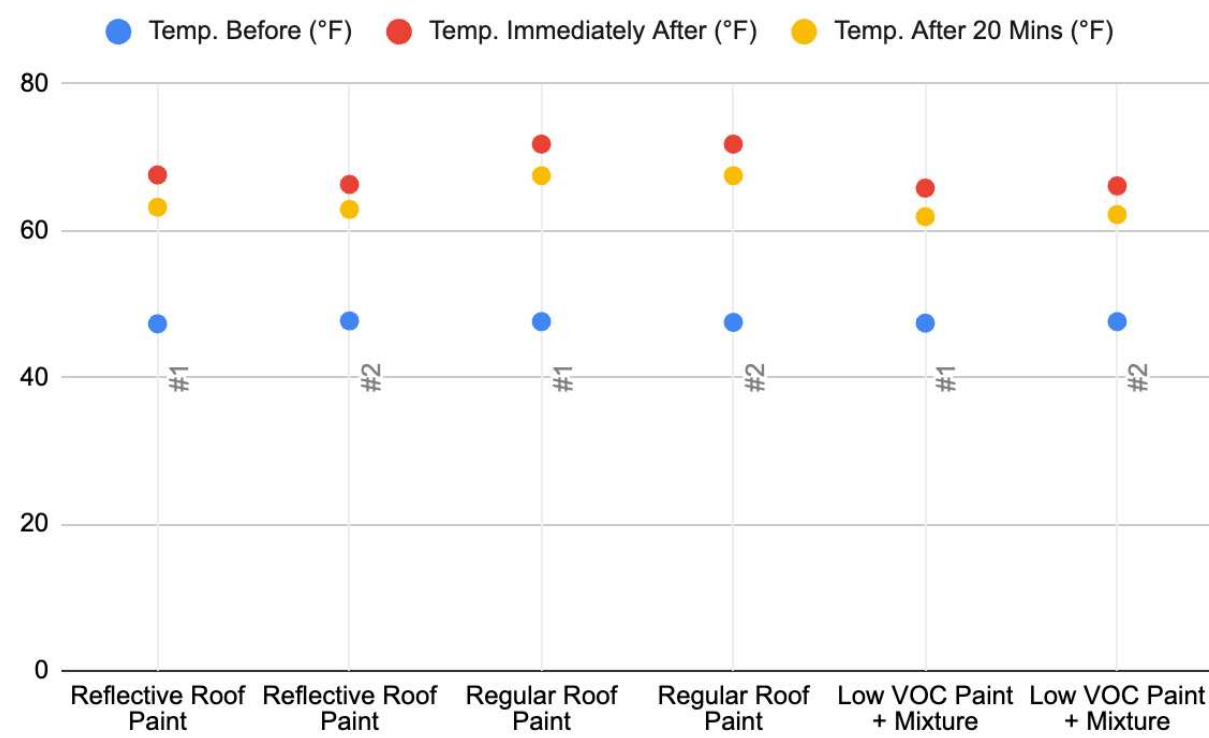


Figure 9. Scatterplot with results

These findings suggest that reflective coatings could help reduce the effects of the **Urban Heat Island effect**, especially in areas with large amounts of pavement and buildings. While the experimental paint has not been fully tested yet, the performance of the reflective paint suggests that similar mixtures using **accessible materials** could also be effective. Overall, this project shows that **low-cost reflective solutions** have strong potential for improving temperature control in urban environments and reducing heat buildup.

Extensions

Future research could test different **pigment concentrations** to improve cooling performance. The experiment could also be done under real outdoor conditions to compare results with natural sunlight. Another extension would be testing coatings on **larger surfaces** and studying their long-term durability. These steps would help better understand how reflective coatings can reduce the **Urban Heat Island effect** in real-world settings.

Citations

MetLink. (n.d.). Urban heat island: Introduction. <https://www.metlink.org/fieldwork-resource/urban-heat-island-introduction/>

National Geographic Society. Urban Heat Island. National Geographic, 19 May 2022, <https://education.nationalgeographic.org/resource/urban-heat-island/>.

NASA. What Is an Urban Heat Island? NASA Science, <https://science.nasa.gov/kids/earth/what-is-an-urban-heat-island/>

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